Categorizing Concepts with Basic Level for Vision-to-Language

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Abstract

Vision-to-language tasks require a unified semantic understanding of visual content. However, the information contained in image/video are essentially ambiguous on two perspectives which manifested on the diverse understanding among different persons and the various understanding grains even for the same person. Inspired by the basic level in early cognition, we propose a Basic Concept (BaC) category that contains both consensus and proper level of visual content to help neural network tackle above problems.

Specifically, a salient concept category is firstly generated by intersecting the labels of the popular ImageNet and the vocabulary of the MSCOCO dataset. Then, according to the observation from human early cognition that children make fewer mistakes on the basic level, the salient category is further refined by clustering concepts with a defined confusion degree which measures the "difficulty" for CNN to distinguish a class pair. Finally, a pre-trained model based on GoogLeNet is produced with the proposed BaC category of 1,372 concept classes. To verify the effectiveness of the proposed categorizing method for vision-to-language tasks, two kinds of experiments are performed including image captioning and visual question answering with the benchmark datasets of MSCOCO, Flickr30k and COCO-QA. The experimental results demonstrate that the representations derived from the cognition-inspired BaC category promote the representation learning of neural network on V2L tasks. Besides, significant improvement gained without modifying standard models.

1. Introduction

For human being, basic level [40] is an abstraction that plays a primary role in the form of early cognition. It contains several objects that are firstly perceived [40] and faster being named by children, then other subordinate concepts are learned afterwards [46]. In the taxonomic hierarchy, basic level is located on the middle which represents mid-level generality (*e.g.* bird, chair), oppose to more general concepts (*e.g.* vertebrate, furniture) and more specific concepts (sparrow, recliner). It also has been found that children make fewer mistakes at basic level on matching and sorting tasks [42, 5] as well as simple decision making on adults. The annotation of ESP dataset [49] is accomplished through an online game which demands two players independently propose labels to one image for matching as many words as possible in a certain time limit. It is demonstrated that humans tend to label visual objects at an easily accessible semantic level termed as the basic level [4]. Similar observations can also be obtained on other human annotated datasets such as image captioning dataset MSCOCO (Microsoft Common Objects in COntext) [25] which limits each description in 16 words. The limitation on time and length will impel people using simple but cognitive useful words for expression.

Vision-to-Language (V2L) are high-level semantic tasks which demand automatically transferring visual information into human language. Recently, more approaches [8, 15, 48] treat V2L as translation tasks which using Convolutional Neural Network (CNN) as the encoder while Long Short Term Memory (LSTM) [10] as the decoder to directly translate an image to a sentence. In the predominant neural network approach, it seems that each partial network has an explicit objective function to learn, however the real evaluation of the generated results is highly subjective which makes V2L challenging both on realization and evaluation.

The main problems V2L facing to are two folds:

1). For different people, the understanding of one image varies.The diversity of thinking habit, attention, and even expression lead to different explanations of visual content. In MSCOCO dataset, each image is associated with 5 captions which are all correct on content but various on expression. This problem makes it difficult to reach a cognitively consensus for same visual content among different persons.

2). For certain person, the understanding of one image can be varied on multiple levels which means it can be very specific (dense caption [14], fine-grained classification [17]) or very general (scene classification [3, 52, 23]). However, for most V2L tasks, especially for image captioning, it only needs a middle level overview of the content. Thus, a proper level of understanding is also vital for effi-

cient representation in V2L.

For human being, these two problems can be largely eliminated once a time by the basic level. Consequently, we are wondering, for neural networks, whether there exists a level resembles the basic level, which possesses the ubiquity and generality of visual representations? Furthermore, could the V2L performance improved by adopting such cognition-inspired mechanism?

Specifically, we firstly generate a salient concept (*SaC*) category that contains candidate basic level concepts by matching ImageNet [41] classes with image captions from MSCOCO dataset. This operation provides a set of cognitively useful words which are located in the middle layers in the ImageNet/WordNet [33] hierarchy. Secondly, we refine these salient concepts by clustering them according to the observation from human early cognition that children make fewer mistakes on the basic level. The confusion degree is defined to measure the difficulty for CNN to distinguish a class pair. Then concepts with small confusion degree are merged to minimize the classification error of CNN. Besides, we visualize examples in the proposed *BaC* and compare it with the basic level in human cognition. Finally, the *BaC* level is used as a semantic representation between vision model and language model in the neural network approach to test its efficiency on Image captioning and VQA tasks.

The main contributions of this works are:

1). We build up a Basic Concept (*BaC*) category that resembling the basic level in human cognition, which not only summarize the consensus among different people but also provide a proper mid-level understanding for V2L tasks. Meanwhile, the proposed categorization procedure also provides a method of automatically generating the basic level for neural networks.

2). The proposed *BaC* level is applied as an optimized semantic representation that conjuncts CNN and LSTM in the standard model. A GoogLeNet [44]model is trained on the *BaC* category and corresponding pre-train model is provided for boosting the training. Significant performance improvements have been achieved by the proposed categorizing method as compared to the corresponding baseline models. For image captioning, the performances in terms of CIDEr [47] and BLEU-4 [37] are improved by 7.7 and 2.1 on the MSCOCO dataset over the baseline. For VQA, the performances have been improved by 2% to 4% against the baseline models.

2. Related Work

2.1. Image captioning

V2L tasks attempt to bridge the gap between vision and natural language, and thus enable universal artificial intelligence to some extent. As compared with other computer vision tasks, V2L tasks such as image captioning [15] and Visual Question Answering (VQA) [28] are more challenging because they require an integrated understanding of visual representation, semantics and natural language.

The purpose of image captioning is to automatically generate natural language descriptions that describe the main content for a given image. Although image captioning achieves significant successes since the wide implementation of deep learning, there are several attempts to provide alternatives to this task. An intuitive approach is the template based methods [18, 9], which detect objects, attributes, actions and scenes, and then put them into a fixed template. However, the generated captions are usually rigid and restricted on expression. Other solutions are based on image retrieval [19, 20], which generate captions by transferring the corresponding descriptions from the retrieved images.

As far as visual question answering is concerned, the task is to provide an accurate natural language answer given an image and a natural language question with free-form about the image content. As compared to image captioning, VQA needs a relatively detailed understanding rather than generic descriptions [1]. Early researches treat VQA as a Turing test proxy such as [28] which generates the answers by the combination of semantic parsing and image scene analysis in a Bayesian framework. More recently, similar to image captioning, end-to-end deep neural networks are employed for VQA [29, 38].

Currently, Convolution Neural Network (CNN) plus Recurrent Neural Network (RNN) is the predominant framework to achieve V2L as image captioning is treated as a machine translation task [2]. Generally, CNN is adopted as the encoder to extract visual features and RNN as the decoder to generate sentences [15, 48]. Between the encoder and the decoder, the CNN features are projected into the same representation space as word embedding to realize mapping from vision to language.

[48, 8] employ the features of the penultimate fully connected layer for visual representation. Besides, recent attention based models [51, 26] utilize the last convolutional layer to obtain spatial information so as to enable the model to adapt "focus" on part of the image during caption generation. These researches prefer to by-pass the elaborate mapping from vision to language. Moreover, the explicit concepts known as attributes are extracted from an additional layer of a carefully pre-trained CNN model in [50], which is adopted for visual feature representation and achieves a significant performance improvement.

2.2. Generic representations

One class of studies focus on producing more human-like recognition results. The majority of these work are naming an image/object by the basic-level concept rather than specific variety in common classification tasks. [6] tries

to balance accuracy and specificity of classification model by high-level concepts if the accuracy is too low. Different from our work, these high-level concepts in [6] are found according to WordNet. Also, [31] uses WordNet structure to find appropriate basic words for the image context instead of generating a "level" of concepts. [34] predicts the entry-level name that people are likely to call from the specific categories located at leaf nodes in WordNet. Similar to [31], it also takes WordNet structure, word frequency, and image classification outputs into account.All these work rely on WordNet structure. However, [35] notice the limitation of WordNet, and believes that visually similar objects are also linked in semantic scenario and introducing visual similarity to basic level extraction.

Besides, [36] highlights that the number of sample in each category can greatly effect the performance of pre-train model. Consequently it suggests to uniform the sample number across classes. For similar reason, [32] combines the full ImageNet dataset to 4k, 8k, 13k categories according to the sample number of each class in WordNet. Trained on this combined dataset, CNNs have a better performance as pre-train models on V2L tasks. [45] introduces human-categorization knowledge to CNN learning process which benefits both classification and V2L tasks.

3. Basic Level of Neural Network

3.1. Categorizing Salient Concepts

Since basic level concepts are objects frequently appeared in daily lives so that children can easily access to during their early learning. It is desired to narrow the range of basic concepts into a smaller semantic concept set which frequently appears in both vision and language.

The benchmark image captioning dataset MSCOCO [25] is employed, which contains 123,287 images and each image is associated with 5 reference sentences describing the corresponding image content. Given an object concept, its most common and visually sensitive substances are included.

Considering that MSCOCO provides only 91 stuffs and obviously it cannot cover as many salient concepts as needed, we intend to align concepts in image caption directly with the annotations in the large-scale image dataset of ImageNet [41]. ImageNet contains 21,841 classes and 14,197,122 images in total, and it is tree structured according to the WordNet organization, which ensures ImageNet has both visual and semantic hierarchical characteristics. Every ImageNet category is corresponding to a WordNet entry which provides rich form and synonyms with the following specific procedure.

Firstly, MSCOCO captions are splitted into words and the frequency of each word is counted. Then, the words with less than 5 occurrences are treated as insignificant and removed. After removing insignificant words, a vocabulary with 9,566 words is constructed and any word outside this vocabulary is replaced by a special "unknown keyword". Afterwards, these filtered 9,566 words are matched to ImageNet annotations and each match is considered as an extracted salient concept. Finally, a Salient Concept (*SaC*) Category is generated which contains 1,689 concepts. We extract a rough set of objects, which contains salient concepts from human annotated datasets.

3.2. Clustering Concepts by Confusion Degree

As is supposed above, the proposed Basic Concept (*BaC*) Category is defined as the superset of basic level concepts. In a semantic hierarchy, the width of basic level is one, then the width of *BaC* is equal or greater than one, so *BaC* is a band that enfolds the basic level. Rather than choosing one strip in the multiple layers, we prefer to aggregate these concepts into a single one.

As aforementioned, basic level is a category where children make fewer mistakes on [42, 5]. So we merge classes by the "difficulty" for CNNs to distinguish it from others. Specifically, we firstly train a CNN until the accuracy reaches its bottleneck, then we measure the "difficulty" according to the error rate of each class. The $\mathcal{S}_{ij}$ is defined as the number of images misclassified from class i to class j. Then for all classes, the directed misclassification among each class-pair can be organized into a confusion matrix $\mathcal{S}$.

We normalize $\mathcal{S}$ by each row and enlarge its partition for better viewing in Figure 1. The long horizontal light indicates the target class that is very prone to be misclassified while the long vertical light means that the corresponding class is visually similar to many other classes. We can make following observations from Figure 1. 1) Most "confusing" classes gathered into groups rather than scattered separately. The orders of classes are arranged by the WordNet hierarchy. Thus, we infer that the visually similar objects are more likely closer in semantic distance. 2) There are more vertical clusters than horizontal clusters. The phenomenon is due to the presence of the container class in the category. Container classes are usually located at the higher of ImageNet hierarchy and contain several sub-classes. For example, "cat" contains "Egyptian cat", "Persian cat", "kitty" etc, and the latter classes are more confused as the former class but not vice versa. 3) There exist several clusters in the same row/column. This phenomenon suggests that some classes that are far away in WordNet may also share similar visual characteristics.

The confusion degree ϕ_{ij} is defined as eq. (1) to measure how difficult a class-pair ij can be distinguished by a certain CNN. The smaller the confusion degree is, the easier for CNN to distinguish a class-pair. This operation also transfers a asymmetric confusion matrix $\mathcal{S}$ to a symmetric similarity matrix Φ.

$$\phi_{ij} = \frac{\mathcal{S}_{ij}\mathcal{S}_{ji} - \mathcal{S}_{ii}\mathcal{S}_{jj}}{\sqrt{(\mathcal{S}_{ii}+\mathcal{S}_{ij})(\mathcal{S}_{ij}+\mathcal{S}_{jj})(\mathcal{S}_{jj}+\mathcal{S}_{ji})(\mathcal{S}_{ji}+\mathcal{S}_{ii})}} \tag{1}$$

It is common that classes which similar to each other have larger ϕ. Merging these classes together is a graph clustering problems. We handle this by the "Graph Distance" d_{ij} defined in eq (2) to metric the graph-based distance between i and j.

$$d_{ij} = \max_{p_{ij} \in P_{ij}} \min_{m,n \in p_{ij}} \phi_{mn} \tag{2}$$

Where P_{ij} indicates all paths from i to j, m, n is an edge in p_{ij} from m to n. The proposed Basic Concept (*BaC*) Category can be obtained by simply merging class i and class j once d_{ij} is larger than a given threshold ϵ.

To decide the value of ϵ, we statistic each confusion degree and the Figure 2 shows the distribution. For most class-pair, their confusion degree range from -0.95 to -1, which demonstrate that CNN can easily distinguish them. However, for other class-pair, CNN can hardly perform well, and the distribution of prone-confused classes is generally uniform. It is worth noting that when ϵ gets smaller its effect of cluster number will be weak since most class-pair own small confusion degree. The selection of ϵ is quite robust from -0.8 to -0.95. In this work, ϵ is empirically set to -0.9 and finally the Basic Concepts category (set of 1,372 concept clusters) is obtained.

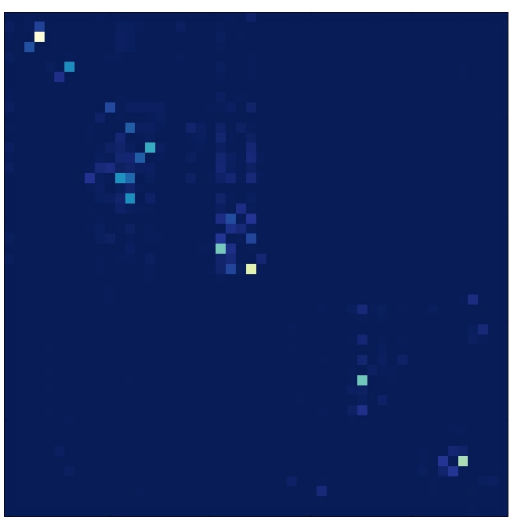

Figure 1. Illustration of Confusion Matrix. The brightness of element ij indicates the number of images belong to class i but being misclassified to class j. The diagonal is removed for better view.

3.3. Comparing BaC with Cognitively Basic Level

The examples are divided into three groups shown in Figure 3: natural object, artificial object, and others. Natural object examples show different varieties belong to the same species and similar geomorphology. Even for the same scenario, various attention will make completely different understandings such as mountain and valley. Artificial object examples, as the large proportions of *BaC* category, mainly includes activities and human created stuffs. The former contains several usually occurrence objects and actions whereas the latter is often various in shape but appears in same scenes. The last group represents examples of social attributes which mainly contains high level semantic inference. All above examples manifest that objects inner one cluster share very general characteristics although their labels vary.

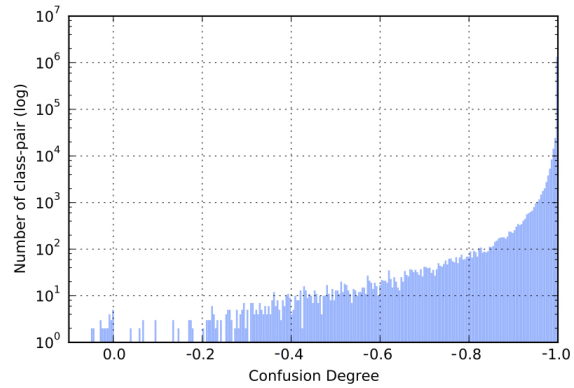

Figure 2. Histgram of Confusion Degree. It shows the number (in logs) of confusion degree for each class-pair in SaC.

Moreover, some partial classes and the corresponding hierarchy related to head dress in WordNet/ImageNet are illustrated in Figure 4, where the words in circle represent the classes in the *SaC* Category (1,689 classes in total) and the words in square indicate the nodes constructed by the hierarchy but not in the *SaC* Category. Besides, the words in the same color represent that the corresponding classes belong to one cluster in the *BaC* Category (1,372 classes in total). Specifically, the words in green and blue in Figure 4 represent clusters with corresponding colors in Figure 3, respectively. Beret, beany and toque have very similar appearances, however they are located on distinct depths and cross with other clusters in the hierarchy.

The visual similarity among such classes can hardly be discovered only by the distance in WordNet. Apparently, visually similar classes may cross the lexical relations among categories. Grouping clusters cannot be accomplished by rolling classes between direct superclass and subclass, or simply selecting a certain layer where all deeper subclasses could be aggregated on. [35] also mention that the encyclopedic knowledge in WordNet dose not coincide with the expected organization of common sense knowledge.

We make following observations by analyzing the results.

1) Natural objects in *BaC* are very similar to the basic level objects in human cognition. We infer the underlying explanation is that human categorizing natural objects essentially depends on shape and texture which are features that neural network good at learning.

2) Artifacts in *BaC* are located relatively deeper in the hierarchy compared with basic level objects. In other words, it is much difficult for NNs (Neural Networks) to summarize the generality among artificial objects as human do, so it inclines to make more clusters on a more specific level. There are two possible explanations of this phenomenon. One is the insufficiency of training data, another is that people categorize artifacts according to function rather than appearance. Consequently, it is harder for NN to infer the latent information for categorization.

4. Experiment

4.1. Image Captioning

Since the BaC level is supposed as a proper level of cognitively consensus, we take it as an optimized representation between CNN and LSTM in a NN approach to verify its efficiency on V2L tasks.

Regarding the image captioning task, the benchmark datasets of MSCOCO [25] and Flickr30k [11] are employed. As mentioned above, MSCOCO contains 123,287 images, and each of them is annotated with 5 captions. The widely adopted split method [15] is applied with 5,000 images for both validation and testing. On the other hand, the dataset of Flickr30k contains 31,783 images and 5 captions for each image. For Flickr30k, 29,000 images are used for training, 1,000 images are used for test, and the rest are for validation.

As far as the evaluation metrics for image description are concerned, the following widely used evaluation metrics are employed for performance presentation, including BLEU [37], METEOR [21], ROUGE_L [24] and CIDEr [47] which are denoted as 'B', 'M', 'R', 'C' for short. About the CNN model for visual representation, GoogLeNet [44] is adopted to encode visual features. First, the GoogLeNet based CNN model is trained with the *SaC*/*BaC* Category by fine-tuning from the 4K pre-trained model [32] for convenience. Then, the model parameters are transferred to the caption task with the following two methods: (1) **OFC** (Original Fully Connected) method, in which the last fully connected layer (i.e., classification layer) of the pretrained model is extracted to be the input to the Long Short Term Memory (LSTM) [10] network. OFC requires that the language model has the same input size and word embedding size as the number of output. (2) **NIC** (Neural Image Captioning) method, which is a commonly used image captioning baseline model [48] with an image embedding layer between CNN and LSTM.

(a) Natural objects

(b) Artificial objects

(c) Others

Figure 3. Toy examples of the BaC category which are divided into three class (a) natural objects, (b) artificial objects, and (c) other examples with latent social attributes.

The caption model follows the default setup of NIC which uses LSTM with 512 memory cells and 512 dimensions of word embedding. The training strategy is also the same as NIC with ADAM optimizer and learning rate is 4×10^{-4}, α is 0.8, β is 0.999, and ϵ is 10^{-8}.

Firstly, the comparisons of different categorizing methods with the same CNN model are demonstrated in Table 1, Table 2, Table 3, Table 4. In these tables, the ILSVRC-1K method [41] is provided by the winner model GoogLeNet with 1,000 official categories of ILSVRC 2012. Methods of Shuffle-4K [32] and Shuffle-8K [32] reorganize the full ImageNet (i.e., 21,841 classes) aiming to solve the data imbalance issue with a subset of classes. The MuCaLe-

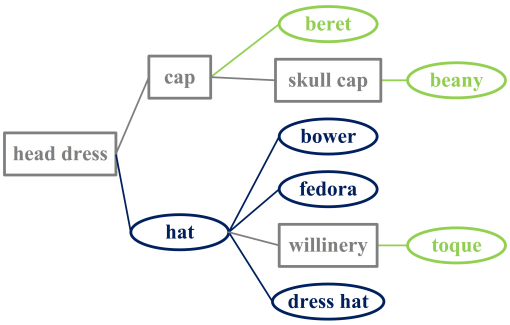

Figure 4. Hierarchical structure of head dress in WordNet. The words in circle represent the classes in the *SaC* Category and the words in square indicate the nodes constructed by the hierarchy but not in the *SaC* Category. Besides, the words in the same color represent that the corresponding classes belong to one cluster in the *BaC* Category. Specifically, the words in green and blue represent clusters with corresponding colors in Figure 3, respectively.

Net [45] contains 480 basic categories from ImageNet-1K (i.e., 1,000 classes)with human annotation.

The comparisons with the OFC and NIC methods on MSCOCO are shown in Table 1 and Table 2, where it can be seen that the proposed *BaC* Category method outperforms all the baseline models in all of the evaluation metrics. When comparing Table 1 and Table 2, the performances achieved by Shuffle-4K and Shuffle-8K with NIC significantly surpass their performances with OFC, and the performances obtained by the proposed *BaC* Category method are stable with both OFC and NIC. The improvement of the *SaC* Category method is obvious with OFC as compared with baseline methods, however, it becomes inconspicuous with NIC. The results imply that the *SaC* Category provides partial salient semantics for word embedding of language model but does not greatly promote the optimization of visual representations, while the *BaC* Category not only enables high-level semantic representation but also presents the generalization of visual features.

Table 1. Comparison of SaC and BaC to other taxonomies of ImageNet by OFC model on MSCOCO dataset.

Taxonomy	C	B-3	B-4	M	R
ILSVRC-1K [41]	90.7	38.7	28.1	24.2	51.9
Shuffle-4K [32]	89.5	38.5	27.6	23.9	51.8
Shuffle-8K [32]	85.9	37.7	26.5	23.3	51.3
MuCaLe-Net [45]	92.3	38.5	27.8	24.2	51.8
SaC Category	91.8	38.9	27.9	24.3	52.0
BaC Category	96.2	40.4	29.6	24.9	52.7

Similarly, the comparisons with the OFC and NIC methods on Flickr30k are shown in Table.3 and Table.4. From the results, it can be observed that both the *SaC* and *BaC* methods are better than the competing baseline methods when employing the OFC method and the *BaC* Category method achieves the best. When considering the NIC method, the proposed *BaC* Category method achieves almost the same as the Shuffle-8K method but with less amount of model parameters.

Table 2. Comparison of SaC and BaC to other taxonomies of ImageNet by NIC model on MSCOCO dataset.

Taxonomy	C	B-3	B-4	M	R
ILSVRC-1K [41]	90.1	37.9	27.6	24.0	51.7
Shuffle-4K [32]	92.9	39.8	29.1	24.5	52.4
Shuffle-8K [32]	92.8	39.0	28.2	24.4	51.9
MuCaLe-Net [45]	92.2	38.9	28.1	24.5	52.1
SaC Category	92.3	38.4	27.8	24.3	52.1
BaC Category	96.0	40.4	29.6	24.7	52.8

Moreover, the comparisons with other state-of-the-art methods on MSCOCO and Flickr30k are shown in Table 5 and Table 6, respectively, where the performance of ROUGE L is not given since almost all of these state-of-the-art methods do not present this result, except Scene+LSTM [13] on MSCOCO in which ROUGE L=50.9 which is lower than 53.8 achieved by the propose *BaC* Category method. Note that the ROUGE L performance achieved by the propose *BaC* Category method on Flickr30k is 44.4. As shown in Table 5, the proposed *BaC* Category method achieves almost the best performance on the criteria of CIDEr and BLEU-4 with its METEOR performance a little bit lower than that of Att+CNN+LSTM [50] . A similar observation can also be made from the comparison result on the dataset of Flickr30k as shown in Table 6, where the proposed *BaC* Category method achieves the best in terms of CIDEr and BLEU-4 while its METEOR performance is a little bit lower than that of Soft-Attention [51] and Hard-Attention [51].

Table 3. Comparison of SaC and BaC to other taxonomies of ImageNet by OFC model on Flickr30k dataset.

Taxonomy	C	B-4	M	R
ILSVRC-1K [41]	34.6	17.8	18.4	42.8
Shuffle-4K [32]	36.2	18.9	18.5	43.2
Shuffle-8K [32]	34.2	18.1	18.0	43.1
SaC Category	36.8	19.4	19.0	43.6
BaC Category	37.4	19.4	19.0	44.0

Several examples of the generated captions by the proposed *BaC* Category method are shown in Figure 5, where the generated captions are given in the first row in blue and the captions generated by the ILSVRC-1K [41] method are also presented in the second row in red for comparison. From left to right, there are three columns indicating the good, fair and bad examples, respectively. The left column shows the captions that accurately describe the objects (*e.g.* trunk) and achieve comprehensive understandings (*e.g.* k-

Table 4. Comparison of SaC and BaC to other taxonomies of ImageNet by NIC model on Flickr30k dataset.

Taxonomy	C	B-4	M	R
ILSVRC-1K [41]	33.1	17.8	18.3	42.6
Shuffle-4K [32]	36.8	18.7	18.8	43.2
Shuffle-8K [32]	36.9	19.0	18.8	43.7
SaC Category	35.2	19.1	18.5	43.6
BaC Category	36.7	19.3	18.8	43.7

Table 5. Comparison of the proposed BaC by NIC model to state-of-the-art methods on MSCOCO dataset.

Method	C	B-3	B-4	M
multimodal RNN [15]	66.0	32.1	23.0	19.5
Google NIC [48]	–	32.9	24.6	–
LRCN-CaffeNet [7]	–	30.4	21.0	–
m-RNN [30]	–	35.0	25.0	–
Soft-Attention [51]	–	34.4	24.3	23.9
Hard-Attention [51]	–	35.7	25.0	23.0
emb-gLSTM [12]	81.3	35.8	26.4	22.7
Jin, et al [13]	83.8	38.1	28.2	23.5
Att+CNN+LSTM [50]	94.0	**42.0**	31.0	**26.0**
VN-Embed [39]	93.7	39.5	29.7	24.4
GLA [22]	96.4	41.7	31.2	24.9
BaC Category (BS3)	**99.7**	**42.0**	**32.0**	25.5

Table 6. Comparison of the proposed BaC by NIC model to state-of-the-art methods on Flickr30k dataset.

Method	C	B-3	B-4	M
LogBilinear [16]	–	25.4	17.1	16.9
multimodal RNN [15]	–	24.0	15.7	15.3
Google NIC [48]	–	27.7	18.3	–
LRCN-CaffeNet [7]	–	25.1	16.5	–
m-RNN [30]	28.0	28.0	19.0	–
Soft-Attention [51]	–	28.8	19.1	**18.5**
Hard-Attention [51]	–	29.6	19.9	**18.5**
emb-gLSTM [12]	–	**30.5**	20.6	17.9
BaC Category (BS3)	**39.5**	**30.5**	**21.2**	18.4

itchen, preparing food rather than table, food). The middle column shows the captions without error but losing some points. As compared with the ILSVRC-1K [41] method, although the proposed method ignores some partial contents of the image (*e.g.*, glove), it produces less incorrect information. Similar conclusions can be derived from the examples with conspicuous errors in the right column. This comparison further verifies that although the language models are the same, the accuracy of visual representation greatly impacts the quality of the generated captions.

4.2. Visual Question Answering

In order to testify the effectiveness of the proposed BaC level on the VQA task, the dataset of Toronto COCO-QA [38] is applied, which belongs to the single-word answer question and contains four types of questions focusing on the object, number, color and location, respectively. We use the official training and test split, which contains 82,783 images for training, 40,504 images for validation and 81,434 images for test, and each has 3 questions and 10 answers. The existing framework for visual question answering is mainly based on the VGG+LSTM model [29], which is also employed as the baseline to implement the proposed *BaC* Category method.

Table 7. Comparison of the proposed BaC to state-of-the-art methods on COCO-QA.

Method	Total	Obj	Num	Col	Loc
GUESS	6.7	2.1	35.8	13.9	8.9
VGG+BoW [38]	55.9	58.7	44.1	52.0	49.4
VIS+LSTM [38]	53.3	56.5	46.1	45.9	45.5
2VIS+BLSTM [38]	55.1	58.2	44.8	49.5	47.3
multimodal CNN. [27]	54.9	–	–	–	–
ILSVRC-1K [41]	54.2	57.5	47.0	50.3	48.5
Shuffle-4K [32]	55.5	57.4	48.3	**52.7**	47.6
Shuffle-8K [32]	56.0	57.7	**48.7**	**52.7**	48.9
BaC Category	**57.2**	**59.3**	48.3	51.2	**51.5**

The accuracy comparison with a number of state-of-the-art methods on the COCO-QA dataset is given in Table 7, where the accuracy criteria of object, number, color and location are presented as "Obj", "Num", "Col" and "Loc", respectively. Additionally, the weighted average accuracy is also listed as "Total" in Table 7. About the competing methods, GUESS is a very simple baseline that predicts the answer only by the question type (object, number, color and location). VGG+BoW [38] performs multinomial logistic regression based on the image feature without dimensionality reduction, and a Bag-of-Word (BoW) vector is obtained by summing all the learned word vectors of the question. In [38], the method of VIS+LSTM uses the last hidden layer of VGG [43] for image embedding and treats it as one word of the question to be encoded at the start and the end of the sentences. The method of 2VIS+BLSTM employs two image features which are put into the language model at the start and the end of the sentences, respectively. On the other hand, the method of multimodal CNN [27] encodes both images and questions by CNN only. Moreover, the methods of ILSVRC-1K [41] and Shuffle-4K/8K [32] are also applied for comparison with the same configuration as the proposed method except the number of BaC categories.

5. Conclusion

For the same image, diverse understandings among various persons and the multiple grained of understanding bring great ambiguity for neural network to manage V2L tasks. To tackle such problems, a BaC category that resembles cognitive basic level is proposed to provide a consensus representation on a proper level. By intersecting classes in ImageNet and MSCOCO, the Salient Concept (SaC) Category

BaC: A group of people standing around a kitchen preparing food.

Baseline: A group of people standing around a table with food.

BaC: A couple of baseball players standing on top of a field.

Baseline: Two baseball players standing on the field with a baseball glove.

BaC: A red and white fire hydrant on a sidewalk.

Baseline: A white and blue fire hydrant with a white background.

BaC: An elephant with its trunk in its mouth.

Baseline: A elephant that is standing in the dirt.

BaC: A wooden bench sitting next to a brick wall.

Baseline: A white and black and white photo of a stone building.

BaC: A statue of a banana sitting on top of a tree.

Baseline: A orange and white orange and a red frisbee in a field.

Figure 5. Comparison of generated captions examples. The left row shows examples without error, the middle row shows examples with minor errors and inappropriate expressions, the right row shows examples with conspicuous mistakes.

is obtained. Then a confusion matrix based method is applied to refine the category thereby providing cognitively more accurate visual representation for language modeling. The comparative experimental results on the image captioning task and visual question answering task have verified that the proposed BaC is able to significantly improve the prediction performance for vision-to-language tasks.

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